Capital, operations and technology - portfolio report

Private review edition | 26 October 2026 | Local-source foundation; external sharing requires approval.

Professional thesis

The portfolio connects private-capital operations, financial and quantitative reasoning, product development and data/technology systems. Management is the mechanism for combining these strands: define a decision, make assumptions explicit, assign responsibility, test failure cases and preserve the evidence needed for revision. The breadth is deliberate. Selective role views condense presentation, not the underlying experience or learning archive. Potential reinforcement is evaluated against a simpler alternative and integration cost; no unique market premium or investment outperformance is claimed.

Evidence boundary

Career narratives are user-attested and source-linked. Project packages combine historical concepts with current AI-assisted documentation and bounded local references. Neither generated documentation nor a course completion proves independent execution. No client deployment, production safety, paid demand or realized economic benefit is asserted. The project evidence separates documentation, model checking, implementation and operational use. The current release uses synthetic/curated fixtures; a real mandate requires fresh input validation and confidentiality review.

Career foundation

BNP Paribas CIB - Private Capital Specialist

June 2025-present

Cash monitoring, reconciliation, transaction booking, ownership-document review and reporting; contribution to document-processing testing within the recorded scope.

Canonical record: EXP-BNP-2025. Verification basis: user attestation; no independent employer confirmation claimed.

Arxa Group / technology-project portfolio

March 2023-March 2025

Project and business development combining research, product definition, financial models and investor engagement. Concept development is distinct from deployment or capital raised.

Canonical record: EXP-PORTFOLIO-2023. Verification basis: user attestation; no independent employer confirmation claimed.

Santander CIB - Corporate Banking Coverage internship

October 2022-March 2023

Financial modeling, market and risk-profile analysis, banking materials and collaboration with relationship, risk and product teams. No final credit-approval authority is implied.

Canonical record: EXP-SANTANDER-2022. Verification basis: user attestation; no independent employer confirmation claimed.

SDG/DGC - infrastructure and renewable-energy project support

October 2021-October 2022

Research, preliminary diligence, business materials, stakeholder coordination and project/executive support in the recorded CEE context. Senior-advisor track record is not attributed as personal delivery.

Canonical record: EXP-SDG-2021. Verification basis: user attestation; no independent employer confirmation claimed.

BCG Platinum Technology Consulting Challenge - finalist

2026

Selected competition achievement, not employment or a client assignment.

Canonical record: ACH-BCG-2026. Verification basis: user attestation; no independent employer confirmation claimed.

Five complementary decision studies

Project	Decision lens	Evidence boundary
Arctervix	Value is not funding feasibility	Bounded agent review; no deployment or market-impact claim
EDNA	Retrieval is not evidence validation	Bounded agent review; no deployment or market-impact claim
Hobbly	Exceptions determine the operating model	Bounded agent review; no deployment or market-impact claim
ARXIS	Traceability does not grant approval	Bounded agent review; no deployment or market-impact claim
ARXA	Prediction has value through a feasible action	Bounded agent review; no deployment or market-impact claim

Qualification architecture

The local inventory preserves 419 course records and 81 program records. These are not 500 independent credentials: program/course overlaps and completion states remain explicit. Representative provider groups provide concise professional views, while the full map retains every source and future option. The archive records 80 programs as completed and one Machine Learning program as pending in its specific record. Vendor-exam preparation is not presented as exam certification. Candidate title mappings are not claims of module-level reading. Selected formal program names and complete grouping rationale remain in QUALIFICATION-GROUPS.yaml and the condensed education block. Choose by target decision context without changing official titles.

Methods and controls

Use source-preserving assumption records, process maps, explicit responsibility, requirements prioritization, scenario analysis, cash-flow reconciliation and metric lineage. The exact module-reading scope is recorded separately; syllabus references do not imply all course content has been reviewed. Each project includes a simpler baseline, a negative case, a bounded recommendation and an account of what the review corrected. The full dossier annexes retain cross-domain reinforcement hypotheses and technical options.

Development options

Private capital/infrastructure: deepen financing, ownership, capital allocation and investment governance. Quant/risk: strengthen temporal validation, uncertainty, model limits and decision constraints. AI/data/product: strengthen evidence systems, data contracts, product economics and operating adoption. These are overlapping views, not mutually exclusive careers. New experience, material, offer or result triggers a review; no calendar review is imposed. One active development path limits work in progress without deleting the option portfolio.

Operational use

A supplied job description produces requirement-by-requirement matching, selected evidence, a private CV/portfolio draft, interview preparation and explicit gaps. A business brief produces alternatives, an operating model, economic assumptions, risks and a decisive test. Missing data yields a provisional assessment, not a precise invented valuation. No internet research, sending, publication or investment action occurs automatically. Source versions and selection decisions are retained; changed or withdrawn evidence requires review before reuse.

Package index

- Arctervix: arctervix-overview.md; arctervix-case-study.md; arctervix-dossier-annex.md.
- EDNA: edna-overview.md; edna-case-study.md; edna-dossier-annex.md.
- Hobbly: hobbly-overview.md; hobbly-case-study.md; hobbly-dossier-annex.md.
- ARXIS: arxis-overview.md; arxis-case-study.md; arxis-dossier-annex.md.
- ARXA: arxa-overview.md; arxa-case-study.md; arxa-dossier-annex.md.

The reports and case studies are private canonical-text views for subsequent PDF, presentation and local web generation. Those formats require their own consistency and visual checks; this text build does not declare them finished.

Arctervix - Value is not funding feasibility

Private portfolio design study. Not a client deployment.

Problem and decision user

A positive investment valuation can conceal an inability to fund the next operating or investment step. The proposed user is investment analyst and capital-allocation decision owner.

Operating and economic approach

Capital allocation and financing decisions under operational and liquidity constraints. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

A positive project NPV can coexist with an immediate cash shortfall. Across 85 synthetic baseline/sensitivity cases, financing and maturity change which fixed schedules remain feasible. Thirteen repository tests cover accounting identities, downside cases and invalid input.

Capabilities illustrated

Cash-flow timing; financing; risk constraints; decision governance. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Annual timing may hide intra-year cash shortfalls; synthetic assumptions cannot establish economic superiority. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study. This specification complements MANAGEMENT-DOSSIER.md and MODEL-VALIDATION-MEMO.md. It does not describe a launched investment service.

Mandate and product boundary

The first product is an internal investment-screening package for an analyst and a capital-allocation decision owner. It answers: which stated investment schedules have positive operating value, which are fundable, and which assumption changes reverse the conclusion? It does not choose a security, estimate an investable strategy's return or issue a financing commitment.

The management proposition is a controlled chain from an economic idea to a reviewable decision. The portfolio engine remains one supporting module. Internal Model connects economic state, observations, institutional dependencies, financing, liquidity, available actions and inference. Only the financing-liquidity-fixed-action slice is exercised in this release. DIM and other original theoretical constructs remain research hypotheses, not validated economic laws.

Required input contract

Input	Type/unit	Control	Missing-data treatment
Mandate and decision owner	Text / stable ID	Identify decision rights and prohibited actions	Pause recommendation
Source record	ID, version, date, confidentiality	Preserve original and content hash	Quarantine unsupported claim
Operating cash flows	Six finite values, EUR thousands	Incremental net operating cash, not accounting profit; disclose taxes and working-capital treatment	No imputed zero
Capital expenditure	Six nonnegative finite values, EUR thousands	Match payment period and scope	No value or feasibility conclusion
Equity	Nonnegative EUR thousands	Treat as opening funding, not operating value	No feasibility conclusion
Debt principal/coupon	EUR thousands / fraction per year	Contractual assumption, not inferred from market availability	Evaluate equity-only explicitly, not as substitute fact
Maturity	Integer years from initial draw	Must remain inside model horizon	Reject unsupported horizon; never shorten silently
Buffer	Nonnegative EUR thousands	Explain reserve policy and operating obligations excluded	No feasibility approval
Discount rate	Fraction per year, greater than -1	Same time/currency basis as cash flows	No NPV assertion
Regime	Normal/restricted	Normal permits one assumed refinancing; restricted prohibits it	Present both scenarios, not invented likelihoods

The reference interface accepts already normalized synthetic inputs. Importing business data requires an analyst to reconcile gross revenue, cost, tax, working capital and capex into those inputs. The current model is not a full company valuation or accounting system.

Process ownership and controls

1. The decision owner defines the mandate and acceptable loss of liquidity. Output: signed-off scope, not a transaction instruction.
2. The analyst prepares source and assumption records, identifies correlated assumptions and constructs the no-investment baseline. A finance reviewer checks units, timing and excluded cash obligations.
3. The model owner runs identical operating assumptions across financing regimes. Output: immutable run with model hash, parameter set, ledger, infeasible actions and sensitivity results.
4. The reviewer reconciles cash and debt identities and reproduces at least one boundary case independently. Until that review actually occurs, the package says agent review, not independent review.
5. The decision owner receives feasible alternatives, counterfactual rejected schedules, downside conditions and information gaps. The decision log records continue/pause/pivot/stop with rationale.
6. A changed source or financing term reopens affected runs and communication outputs. Historical versions remain available; no dashboard may silently retain old numbers.

Roles are proposed functions. The current evidence does not establish that a staffed investment committee or client has used this workflow.

Architecture and sourcing choices

Layer 1 preserves source documents and assumptions. Layer 2 normalizes units and time conventions. Layer 3 computes separate operating value and financing ledgers. Layer 4 validates identities, limits and scenarios. Layer 5 produces the memo, dashboard and portfolio case. IDs and hashes connect these layers.

Use local Markdown/YAML and Git for controlled narrative and models; a controlled spreadsheet is the intended analyst-facing calculation view. Python is retained only as the independent deterministic reference and regression check. Existing collaboration and presentation products can be reused after access and confidentiality review. No SaaS connection, paid license or production integration is assumed.

A future data adapter must preserve source timestamps, missingness, currency, period grain and revision history. A future optimizer must first demonstrate improved decisions against these fixed rules, net of modeling, review and maintenance cost. ML, programmable contracts and blockchain remain optional; none resolves an identified first-release need.

Service economics and adoption

No demand, revenue or willingness-to-pay estimate is claimed. An internal pilot can measure analyst preparation time, reviewer correction time, rejected invalid inputs and decision turnaround. Loaded hourly costs, data fees and maintenance must be supplied before estimating delivery economics. Compare the integrated workflow with an NPV-only sheet and a manual financing review using the same cases.

Potential reinforcement is conditional: liquidity controls may prevent a superficially attractive but unfinanceable recommendation. The cost is extra input collection and review. The release demonstrates this mechanism on synthetic schedules; it does not prove a net economic benefit in an organization.

Risk register and stop rules

Risk	Proposed owner	Control	Residual limitation / escalation
Annual aggregation hides shortfall	Finance reviewer	Require a finer cash calendar before real commitment	Current six-period model cannot approve intra-year liquidity
Optimistic common assumptions bias both value and liquidity	Decision owner	Downside cash/capex cases and explicit source challenge	No probabilistic calibration
Refinancing unavailable	Treasury/finance function	Restricted regime; confirmed terms required for use	Synthetic regime not a credit offer
Funding counted as value	Model owner	Unlevered NPV excludes equity, draw and refinancing	Equity return is not calculated
Invalid input or silent horizon change	Model owner	Finite-value checks and explicit maturity rejection	No automatic horizon extension
Confidential sources in presentation	Publication reviewer	Allowlisted claims; no source-document export	Requires separate approval
A polished study mistaken for deployment	Portfolio owner	Maturity labels in every format	No client, revenue or performance claim

Pause if required source, owner or financing term is unavailable. Pivot if a simpler review achieves the same decision quality at lower total cost. Stop an investment schedule that breaches its required buffer; this does not automatically stop the research project.

Acceptance and release evidence

ARCTERVIX-REQ-01 is checked by ledger identities and the financing-independent NPV regression test. ARCTERVIX-REQ-02 is checked by explicit infeasible status, first breach and the memo's exclusion rule. ARCTERVIX-REQ-03 is supported by model/source hashes and archive change-impact checks; a change requires a new reviewed run rather than an assertion of automatic economic correctness.

The bounded review covers 16 baseline and 69 sensitivity cases, plus 13 repository tests. It includes no refinancing, earlier maturity, delayed operating flows, negative operations, insufficient cash, capex changes and invalid inputs. The evidence supports the design-study claims only. Production security, regulatory suitability, external demand and observed business outcomes are not acceptance claims of this package.

Interview narrative and next options

Problem: a positive project valuation can hide a funding constraint. Decision: separate operating value from the cash and debt path before considering adaptive optimization. Difficulty: timing conventions and horizon boundaries can silently change the financial question. Correction: reject maturity outside the modeled horizon and non-finite inputs; expose counterfactual flows after a breach. Result: a reproducible synthetic distinction between value and feasible action. Limit: not a calibrated real-options price or an observed investment achievement.

Next options are a working-capital bridge, finer cash timing, empirical assumption review or an adaptive policy comparison. Activate an option only for a concrete mandate and evidence gap. Preserve broader technical knowledge without making every possible technology a requirement.

Validation evidence and reflection

Private design-study evidence. All monetary values are synthetic EUR thousands; years 0-5. Not an investment recommendation.

Decision

Retain the fixed-policy ledger as a screening tool. Separate project value from funding feasibility. Reject schedules after a buffer breach; do not rank their later counterfactual cash flows as achievable returns. An adaptive optimizer is not justified by this test.

Baseline results

Policy	Funding	Regime	Project NPV	Minimum cash	Buffer gap	Status
invest	equity	normal	13.72	-20.00	30.00	infeasible
invest	equity	restricted	13.72	-20.00	30.00	infeasible
invest	mixed	normal	13.72	20.00	0.00	feasible
invest	mixed	restricted	13.72	20.00	0.00	feasible
stage	equity	normal	2.68	-22.00	32.00	infeasible
stage	equity	restricted	2.68	-22.00	32.00	infeasible
stage	mixed	normal	2.68	14.80	0.00	feasible
stage	mixed	restricted	2.68	6.60	3.40	infeasible
defer	equity	normal	5.40	-25.00	35.00	infeasible
defer	equity	restricted	5.40	-25.00	35.00	infeasible
defer	mixed	normal	5.40	15.00	0.00	feasible
defer	mixed	restricted	5.40	15.00	0.00	feasible
reject	equity	normal	0.00	80.00	0.00	feasible
reject	equity	restricted	0.00	80.00	0.00	feasible
reject	mixed	normal	0.00	80.00	0.00	feasible
reject	mixed	restricted	0.00	80.00	0.00	feasible

Prior forecast and observed model result

The archived agent forecast is preserved unchanged. H-IM-01, H-IM-02 and H-IM-03 all match the reference calculation: {'H-IM-01': True, 'H-IM-02': True, 'H-IM-03': True}. Stage with restricted refinancing is infeasible at maturity 1 (minimum cash -25.20) and feasible at maturity 4 (minimum cash 14.80). These are deterministic implications of selected assumptions, not estimated population effects. The user forecast remains separate and optional.

Sensitivity and decision boundaries

Reviewed 69 sensitivity scenarios plus 16 baselines. One input changes at a time: discount rate, cash buffer, initial equity, coupon, maturity, capex scale or one-year operating delay. Full ledgers are in artifacts/management-review.json. The delay scenario drops cash flows beyond year 5; it combines timing loss with a shorter observed operating horizon, not a perpetual loss estimate. A higher buffer can remove financing feasibility without changing unlevered NPV. More equity can restore feasibility without proving that the extra capital earns an attractive return. Higher capex can impair both value and liquidity. Refinancing availability changes financing cash, not the operating cash-flow schedule.

Operating handoff

Decision owner: proposed investment lead. Model preparer: proposed analyst. Independent check: finance/risk reviewer; no staffed team is claimed. Intake requires dated operating assumptions, capex schedule, funding commitments, maturity, coupon, buffer and unit conventions. The reviewer reconciles cash and debt identities, checks timing and compares an explicit no-investment baseline. Escalate any missing commitment, horizon mismatch, negative liquidity or unreviewed source change. No capital commitment follows automatically.

Source application

Local Wharton C419-193 transcripts on time value and alternative scenarios support time-indexed discounting, what-if analysis, break-even reasoning and the need to check working capital and capacity. They do not validate the synthetic parameter choices or establish empirical financing effects. Source-reading details and hashes belong to the archive reading register. Future real inputs must identify nominal/real basis, taxes, working capital, within-year timing and terminal treatment before a consequential valuation.

Post-mortem

Review exposed silent maturity clipping and acceptance of non-finite inputs. Both were replaced with explicit errors and regression tests. This improves input control; it does not make the model production-ready. Alternative explanation: the observed financing difference follows mechanically from the selected cash buffer and repayment dates. A different operating schedule or initial capitalization can reverse the feasible set. Bounded conclusion: the model is suitable to explain the distinction between value and feasibility under stated assumptions. It is not a calibrated real-options price, optimal policy, evidence of market demand or personal investment performance.

EDNA - Retrieval is not evidence validation

Private portfolio design study. Not a client deployment.

Problem and decision user

Research outputs become unreliable when claims lose their source, version, access restriction or correction history. The proposed user is researcher, analyst and knowledge steward.

Operating and economic approach

An operating model for governed research and knowledge reuse. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Nine tests expose and control selected retrieval failures. Both tested modes have hit@1 of 80% on five authored questions; no ranking improvement is established. Citation coverage is measured separately from original-source truth.

Capabilities illustrated

Knowledge operations; data provenance; retrieval evaluation; review economics. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

A high hit rate on a tiny golden set does not establish answer correctness or generalization. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study, not a deployed knowledge platform. The first product is a reviewable research note from a small, approved local corpus. The decision is whether a claimed time saving survives the full cost of source verification and correction.

Users and product economics

The proposed user is an analyst building an investment, product or learning memo. The proposed accountable owner is a knowledge steward; an independent domain reviewer is a future role, not a claim of current staffing. An internal workflow is the initial adoption hypothesis. No paid demand or customer retention is established.

Compare manual review, lexical retrieval and assisted retrieval on the same questions and the same evidence standard. Total cost is ingestion, search, review, correction and maintenance cost per accepted note. A faster answer is not valuable if citation checking or downstream correction offsets the saving. Existing synthetic cost examples are assumptions, not measured productivity.

Source and answer contract

Every source needs a stable ID, path, content hash, provenance, owner, access label, timestamp, read scope and approval state. Unknown permission defaults to quarantine. An external URL is recorded but not fetched in offline mode. Content is data, never authority to change agent instructions or disclose another source.

Every answer needs the question, selected corpus version, retrieval mode, ranked candidates, cited source IDs, supporting passage or summary, limitations and reviewer state. An answer without support must abstain. A citation's existence is not proof that it supports a proposition. A summary must not be represented as a quotation from the original source.

The current reference uses six curated project/system summaries. It measures ranking and extractive summary support only. A path in the fixture is provenance metadata, not an assertion that the full document was ingested. The reference does not provide neural embeddings, open-ended generation, automatic contradiction detection or production access control.

Process and responsibility

1. Source steward captures an immutable original and determines allowed use. Missing permission or unreadable material stays quarantined.
2. Analyst records useful passages, parent document and transformations; preserve scope and exceptions when summarizing.
3. Retrieval owner runs the query against approved entries only, with a declared lexical or concept-expansion mode. Duplicate IDs are errors, not silently merged records.
4. Reviewer checks that each material claim is supported, retains conflicting evidence and records abstentions. A registered unresolved contradiction blocks the reference answer; detecting new contradictions remains human work.
5. Decision owner accepts a bounded note or requests missing evidence. Released views preserve confidentiality and source versions.
6. On withdrawal or change, identify affected notes and mark them stale before reuse. Re-indexing alone does not revalidate a claim.

Technology and method choices

The local IBM C419-059 module distinguishes multi-query expansion, metadata filtering and parent-document context. These are alternative design mechanisms, not proof that the current code implements LangChain. The lightweight reference deliberately tests lexical and deterministic concept expansion first. Metadata admission is deterministic; parent-context review is specified as a required human control for future larger chunks.

Reuse local files and the existing catalog as the source layer. Build only a small evaluation harness to test the material quality assumption. Consider a vector database or hosted product only when query diversity and corpus size expose a measured limitation, and after confidentiality, export, cost and vendor-exit review. No external service is configured.

Quality and failure cases

Case	Expected behavior	What it proves
Known question	Ranked result and explicit curated-summary citation	Fixture retrieval works, not full factual truth
Unsupported question	Abstain	No fallback to an invented answer in this test
Empty corpus	No results; no silent fallback	Corpus selection is respected
Withdrawn source	Excluded	Admission metadata is applied
Duplicate IDs	Reject input	Identifier ambiguity is visible
Registered contradiction	Abstain and expose conflict ID	Manual conflict registration affects reuse
Source revision	Package becomes stale	Integrity control, not automatic semantic correction

Dashboard metrics: hit@k and reciprocal rank for retrieval; cited-answer share and exact-summary support for fixture evidence; unsupported-answer rate, contradiction handling, abstention and total review minutes for a real evaluation. Each denominator and corpus version must be recorded. Do not combine fixture percentages with measured user productivity.

Handoff, stop rules and narrative

Deliver the corpus card, access decisions, query set, expected sources, test output, failure log, cost assumptions and memo to a domain reviewer. Pause if permissions, parent context or a material contradiction are unresolved. Pivot to manual review when the assisted process costs more at the same evidence quality. Scale only after a recipient confirms useful accepted notes.

Interview narrative: the problem is not producing fluent text but preserving evidence through a research workflow. The design separates retrieval from support and approval. Review found an inaccurate TF-IDF description, hard-coded citation coverage and an empty-corpus fallback. Correcting these is a documented quality-control result; it is not a commercial deployment or proof of reduced hallucinations in arbitrary language models.

Limitations: tiny hand-selected corpus, English tokenization, correlated evaluation examples, no neural retrieval, no measured human review time, no automatic causal or truth inference. These limitations constrain the recommendation without discarding future AI/data options.

Validation evidence and reflection

Private bounded design study. Five authored retrieval questions, six curated summaries and synthetic cost assumptions; no external market or user trial.

Evidence quality

Both lexical and concept-expansion modes achieve hit@1 80%, hit@3 100% and MRR 0.90 on this fixture. There is no demonstrated ranking improvement. Measured citation coverage is 80% for lexical and 100% for concept expansion. Exact-summary support has the same rates. This measures extracted summary support, not correctness against the original documents or relevance of every answer. The original no-worse-at-three prediction passes, but it is a weak acceptance condition and does not justify investing in a semantic platform. Empty corpus, unsupported query, withdrawn source and registered conflict all produce abstention. Duplicate IDs and unsupported modes are rejected in regression tests. Novel contradictions are not detected automatically.

Economics and decision boundary

At the declared synthetic cost assumptions, manual review costs EUR 60.00 per accepted note; assisted preparation plus review and tools costs EUR 42.00. With 60 review minutes the assisted cost becomes EUR 87.00. The break-even review time is 33.00 minutes at the same acceptance quality. Time beyond that threshold eliminates the assumed saving. Ingestion and maintenance are not measured; allocate them before claiming net benefit. The pre-recorded cost forecast matches arithmetic, not observed productivity. A higher cited-answer rate can simply mean less abstention; it is not automatically a quality improvement.

Recommendation

Continue with a controlled local evidence workflow and explicit review; do not scale or buy a hosted retrieval service on this result. For the next supplied corpus, define approved passages and expected answers before running the comparison, record manual and assisted review time, and include stale, conflicting and out-of-scope questions. Pause a note when required evidence or permission is missing. Prefer manual review if the full assisted cost exceeds it at the same quality. Preserve advanced retrieval techniques as options conditional on a demonstrated bottleneck.

Post-mortem and handoff

Corrected the inaccurate TF-IDF label, hard-coded citation coverage and empty-list fallback. Added explicit source-summary scope, excerpt hashes, withdrawn-source filtering and conflict abstention. These changes improve auditability within this narrow reference. Handoff: DELIVERY-SPECIFICATION.md, MODEL-CARD.md, core.py, tests and artifacts/management-review.json. Roles and external integrations are proposed. No original course material is copied into an outward-facing case study.

Hobbly - Exceptions determine the operating model

Private portfolio design study. Not a client deployment.

Problem and decision user

A reservation is not operationally complete until capacity, payment, cancellation and settlement states agree. The proposed user is activity organizer and customer.

Operating and economic approach

A booking-service operating model with explicit transaction economics and exception ownership. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Eight tests cover selected booking/payment states, conflicting request keys, cancellation, full refund and reconciliation. Synthetic contribution changes from EUR 2 to EUR -0.50 per booking when support cost rises.

Capabilities illustrated

Product definition; transaction operations; exception ownership; unit economics. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Synthetic payment state tests do not establish provider integration, fraud prevention or legal compliance. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study. The product hypothesis is a booking service that connects organizers and customers while making exceptions, payment states and service costs visible. No live payment integration, customer base or regulatory authorization is claimed.

Business and product model

Organizer: publishes an offering, price and capacity, confirms service delivery and handles organizer-side cancellation. Customer: selects availability, books, pays and requests support or cancellation. Platform operations: reconciles states and exceptions. A payment provider would execute money movement under a separately agreed commercial and compliance model. Platform custody of funds is not assumed.

Potential revenue is a fee or take rate, not gross booking value. The reference economic case uses a synthetic EUR 50 booking and 10% take rate, giving EUR 5 platform revenue. Deduct EUR 1 payment cost, EUR 1.50 support and EUR 0.50 expected loss: EUR 2 contribution before fixed costs and tax. At EUR 4 support cost the contribution becomes EUR -0.50. These are assumptions, not observed transactions or a pricing recommendation.

Alternatives: manual scheduling plus an existing payment link; an existing booking product; a custom integrated service. Begin with the simplest approach that meets capacity, cancellation and reconciliation requirements. A new platform is justified only when differentiation and net economics survive a real operational comparison.

State and responsibility contract

Event	Required state/control	Result	Proposed owner
Publish listing	Integer minor-unit price, positive capacity, organizer mandate	Bookable offering	Organizer
Reserve	Listing exists; capacity available	Pending payment, slot held	Booking service
Capture	Pending payment; unique request key	Confirmed booking and captured amount	Payment orchestration
Payment failure	Valid simulated failure outcome	Payment failed; hold remains until cancelled	Support/operations
Cancel	Pending, failed or confirmed booking	Cancelled; capacity released	Authorized service operator
Full refund	Cancelled booking with captured payment	One refund of captured amount	Payment operations
Replay request	Same key, same booking and outcome	Same result; no second capture/refund	Payment orchestration
Conflicting key	Different booking/outcome	Reject; investigate	Operations

The reference has no hold timeout, customer authentication, partial refund, retry-after-failure workflow, authorization/capture split, settlement schedule or chargeback process. Those are specified next-design decisions, not silently implied functions. Production roles must enforce access rights; named owners here are proposed functions.

Reconciliation and exception handling

Keep booking, payment, refund and settlement records separate. Gross captured less gross refunded is a payment-flow measure, not revenue or profit. Reconcile provider events to internal request IDs, amounts, currency and state; unresolved differences remain an exception queue with owner and timestamp. Never mark a refund paid solely because an internal request was accepted.

The local reference simulates capture and full refund and checks their aggregate identity. It does not reconcile a real provider statement. Concurrent requests require transactional reservation, unique event processing and race-condition tests before operational use. A support agent must not manufacture a successful payment state to resolve a customer complaint.

Data and technology design

Listing: ID, organizer, product terms, slot/capacity, price in minor units, currency and cancellation version. Booking: ID, customer reference, listing/slot, immutable accepted terms, timestamps and status. Payment event: provider event ID, request key, booking, amount, currency, status and audit time. Refund: original payment reference, authorized reason and amount. Avoid storing card credentials; use a suitable provider boundary after separate review.

The current reference stores a deliberately smaller synthetic subset in SQLite. Currency is a fixed demonstration assumption rather than a multi-currency field. Local documentation is canonical; UI,

workflow board and dashboard are controlled views. Build/buy review must include transaction fees, refund treatment, payout schedule, support cost, integration effort and vendor exit-not only developer effort.

Dashboard and management decisions

Show booked versus fulfilled volume separately; capture/refund/reconciliation amounts separately; contribution per completed booking with support and expected-loss assumptions; unresolved exception age and owner; failed-payment and refund rates with denominators. Missing settlement data means unknown, not zero. Refresh on new events or revised assumptions, and retain prior snapshots.

At the synthetic base assumptions support cost must stay below EUR 3.50 per booking for positive contribution before fixed costs. This is an arithmetic boundary, not proof that it is operationally achievable. The simpler alternative may win after acquisition, organizer onboarding, disputes and fixed costs are included.

Risks, validation and handoff

Material risks: overselling under concurrency; duplicated or misattributed payment requests; cancellation/refund disagreement; uncosted support; mistaken revenue recognition; access/privacy failures. Controls: atomic inventory reservation, request fingerprinting, provider reconciliation, exception ownership and source-backed metrics. The small reference validates only selected sequential state transitions and arithmetic.

Handoff includes the state table, API/event contracts, refund policy decisions, synthetic tests, economic assumptions and a list of unimplemented controls. Pause deployment until the payment/compliance boundary and operational ownership are approved by suitable specialists. This is not a statement of compliance with any jurisdiction's current law.

Interview narrative: connect product economics to the exceptions a service must actually operate. The review found cross-booking idempotency risk and missing refund/reconciliation coverage. The corrected reference demonstrates those controls locally. It does not demonstrate a production marketplace, revenue or customer adoption.

Validation evidence and reflection

Private synthetic study. Eight repository tests cover capacity, key replay/conflict, valid states, cancellation, refund, amount reconciliation and invalid inputs.

Result

One capture of 5000 cents and one full refund of 5000 cents reconcile to zero net payment flow. Replay produces no second financial event. Cross-booking key reuse is rejected. Cancellation releases the local slot. This is a sequential local reference, not provider settlement.

At the separately declared economic scenario, platform contribution is EUR 2.00 per completed booking and EUR -0.50 when support cost rises. The break-even support cost is EUR 3.50, before fixed costs and tax. Gross booking value is not platform revenue. The pre-recorded cost forecast matches the arithmetic.

Decision

Continue documenting the service and compare a manual/existing-product baseline. Do not deploy a payment service from this reference. Confirm cancellation terms, support workload, refund fees, settlement rules and access responsibilities before implementation. A favorable unit contribution does not establish total profitability.

Post-mortem

The earlier implementation replayed a key without checking the booking identity and treated any unknown outcome as failure. Both now reject ambiguous input. Full-refund reconciliation was absent and is now demonstrated. Authorization/capture, concurrency, partial refunds and chargebacks remain explicit gaps; no claim of production safety is made.

Source and handoff: DELIVERY-SPECIFICATION.md, MODEL-CARD.md, the local reference tests and artifacts/management-review.json. The historical technical demo uses a different synthetic pricing scenario; do not combine its numbers with this management scenario.

ARXIS - Traceability does not grant approval

Private portfolio design study. Not a client deployment.

Problem and decision user

A changed requirement can leave inconsistent decisions, versions and deliverables in circulation. The proposed user is design lead, documentation coordinator and project reviewer.

Operating and economic approach

An operating model for traceable design documentation and controlled project change. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Seven tests check every decision-to-artifact obligation, duplicates, unknown references and version/actor controls. A matching version permits human review, not automatic engineering approval.

Capabilities illustrated

Requirements; document governance; change control; cross-functional coordination. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Traceability completeness does not prove correctness of engineering design or compliance with building law. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study for one synthetic residential project. The product is a controlled documentation workflow, not a CAD/BIM/GIS integration or an engineering approval service.

Problem, customer and alternatives

A project manager needs to know which requirement caused a decision, which artifact implements it, who may approve a change and which downstream work must be reconsidered. A folder of drawings alone does not answer that question. The proposed users are the project manager, discipline leads, document controller and client representative. These are role designs, not a staffed organization.

Alternatives are a controlled spreadsheet/register, an existing common-data environment, or a custom traceability layer. Prefer an exportable existing tool unless a concrete cross-document dependency cannot be handled. No paid demand, procurement outcome or reduction in construction errors is established.

Information contract

Requirement: ID, statement, source, owner, acceptance criterion and status. Decision: ID, requirement links, alternatives, rationale, approver, date and effective version. Artifact: ID, decision links, version, author, checker, approval state, confidentiality and content hash. Change request: affected requirement, reason, expected source versions, cost/time estimate, reviewer, authority and disposition.

The synthetic fixture names three illustrative documents; these names are not actual architectural drawings or certified engineering reports. Rule-based document classification is only a suggestion requiring review; it is not a trained AI classifier or technical compliance check.

Operating workflow

1. Project manager captures the client need and mandate. Discipline lead translates it into reviewable requirements; document controller registers IDs and versions.
2. Analyst prepares alternatives. The authorized decision owner records the selected option, unresolved assumptions and acceptance criteria.
3. Author produces an artifact. Checker confirms every linked decision is addressed; one artifact somewhere under a requirement is not sufficient if another decision remains uncovered.
4. A change request identifies the expected versions before review. The reference rejects unknown actors, flags version mismatch and returns affected decisions/artifacts. It never grants engineering approval.
5. Discipline leads assess cost, schedule, constructability and adjacent requirements. Project manager consolidates dependencies. The authorized owner approves, rejects, pauses or requests redesign.
6. Document controller issues the new baseline, preserves superseded versions and informs affected recipients. Unreviewed downstream documents become stale, not silently current.

Economics and constraints

The synthetic management case assumes four coordination hours at EUR 60 plus EUR 10 allocated tooling cost: EUR 250 per change. Eight hours cost EUR 490. These are coordination costs, not project savings or total construction change costs. Materials, contractor variations, delay penalties and design liability are excluded.

A change memo must separately show direct implementation cost, review effort, schedule impact and uncertainty. Missing cost or date information pauses a definitive recommendation. Compare improved traceability against its data-entry and governance burden; extra fields are justified only when they change a decision or prevent a specific error.

Architecture and tool boundaries

Canonical requirements and decisions remain in exportable records; diagrams and task boards are views. A document platform must support immutable issued versions, access control and change history. The local reference checks graph structure and explicit version/actor inputs only. An approver name passed to a function is not authentication or secure authorization. Real deployment requires identity integration, immutable audit logging and specialist review.

Use the course-derived iterative requirements approach to clarify user stories, update the backlog and review a changed need with stakeholders. This system remains event-driven; the course's sprint cadence is not an imposed calendar requirement. Existing architecture/construction materials provide candidate domain extensions, not permission to claim professional certification or code compliance.

Dashboard, risks and acceptance

Metrics: requirements with all decisions represented; orphan/dangling links; stale artifacts; pending approvals by owner; coordination cost and age of unresolved changes. Every denominator uses a declared project baseline. Completeness is a structural property, not proof that the design is safe or correct.

Negative cases: duplicate IDs, unknown references, an uncovered second decision, unauthorized actor, old expected version and missing artifact. Hand off the fixture, dependency matrix, change assessment, limitations and review memo. Pause when authority or baseline is unclear. Stop a custom-tool build if an existing controlled register meets the same needs more cheaply.

Interview narrative: the main contribution is separating traceability from approval. Review exposed a false-completeness case where one artifact hid another uncovered decision. Corrected checks and version-aware impact assessment make the governance question explicit without claiming engineering validation.

Validation evidence and reflection

Private synthetic study. Seven repository tests check traceability, missing artifacts, duplicate/dangling IDs, uncovered decisions, rule-based classification disclosure and actor/version controls.

Result and forecast

The complete three-requirement fixture has structural coverage; removing an artifact or adding an uncovered decision exposes a gap. A version mismatch pauses the change; a recognized actor with matching versions is ready for human impact review, never automatically approved. The original completeness/negative-case prediction holds within this authored fixture.

The pre-recorded coordination-cost case gives EUR 250 per change at four hours and EUR 490 at eight hours. This is assumed effort cost, not a measured saving. Doubling review time does not double the fixed EUR 10 tooling allocation.

Decision and limitations

Continue with a controlled requirements/decision/version register and a human change-review workflow. Do not infer engineering correctness from graph completeness or deploy the actor-input check as an access-control system. Obtain actual scope, approver mandate, cost and schedule implications before issuing a change.

Post-mortem

Earlier completeness checked whether any artifact existed beneath a requirement; it could conceal an uncovered second decision. The revised check requires every linked decision to be represented. Duplicate and dangling references are rejected. The reference classifier remains deterministic keyword matching, not a trained AI model. A controlled spreadsheet or existing document system may meet the need without new software.

Handoff: DELIVERY-SPECIFICATION.md, source fixture, tests and artifacts/management-review.json. No client adoption, construction saving or regulatory compliance is claimed.

ARXA - Prediction has value through a feasible action

Private portfolio design study. Not a client deployment.

Problem and decision user

An industrial analytics initiative can optimize a prediction metric without improving the operational decision or economics. The proposed user is operations manager, asset owner and finance sponsor.

Operating and economic approach

A model for managing data-enabled operational transformation and investment choices. The package joins the business question, product scope, operating handoffs, technology/data assumptions and financial consequences. Existing tools and simpler alternatives remain part of the decision.

Checked result

Eight tests and twelve efficacy/capacity scenarios show how intervention assumptions can reverse a favorable cost comparison. The act/defer calculation is a cost comparison, not a real-option price.

Capabilities illustrated

Industrial analytics; action constraints; economic scenarios; staged investment. These are demonstrated documentary/model mechanisms, not a certification of personal independent mastery.

Limits and next use

Distribution shift, sensor quality and intervention effectiveness may invalidate estimated savings. Use the detailed specification and validation memo to scope a real mandate, identify missing data and define the smallest decisive test. Publication and deployment require separate review.

Delivery specification

Private design study. The first product is a decision package for one maintenance intervention, not a general industrial AI platform. The economic question is whether an executable action improves expected operating cost after intervention, capacity, data and integration costs.

User, value and alternatives

Proposed users: asset/operations owner, maintenance planner, data steward and finance reviewer. A forecast matters only if an authorized operator can take a timely, safe action. The adoption hypothesis is an internal decision-support pilot; no factory deployment, paid demand, causal maintenance effect or realized saving is established.

Compare no intervention, a simple fixed rule, forecast-triggered action and a cost/capacity-constrained policy. A more accurate score is not automatically a better decision. Prefer an existing maintenance-management process where the additional data/model cost exceeds the incremental benefit.

Data and model contract

Required real inputs: asset identity, timestamp/grain, sensor units/calibration, maintenance history, failure definition, exposure time, missingness, downtime cost, action cost, lead time, resources and safety constraints. Preserve temporal ordering and distinguish observations before and after intervention. Do not train or evaluate on future information.

The reference generates 80 deterministic periods. Its risk score is an engineered index, not an empirically calibrated failure probability. The rolling forecast predicts that index; failure labels are generated by a separate deterministic rule. Forecast MAE therefore does not measure calibrated failure risk. This fixture is a mechanism demonstration and must not support an investment return claim.

For new data, reconcile labels and units, establish a temporal holdout, define drift/missingness controls and compare with a simple baseline before estimating business value. Unknown intervention effectiveness remains an uncertainty scenario, not 100% by default in a real decision.

Operating process and ownership

1. Operations owner selects one asset, failure mode and decision window; finance owner records material costs and acceptable exposure.
2. Data steward checks timestamps, units, missing records and measurement changes. Unsafe or unexplained data pauses the model-assisted recommendation.
3. Analyst forecasts only from available past observations and documents uncertainty. Model owner compares action policies using the same external cases.
4. Maintenance planner checks crew, spare parts, lead time, production windows and capacity. A predicted failure without an executable intervention is an escalation, not a successful model outcome.
5. Authorized operator decides; safety constraints override the score. Log action, reason and observed result separately from forecast.
6. Reviewer compares predicted and observed outcomes, including false alarms, missed failures, intervention failures and opportunity cost. New evidence changes the policy or pauses it.

Economics and financing

Separate the historical technical fixture's large synthetic cost units from the management case in EUR per period. The management case assumes failure probability 0.10, failure loss 1000, two interventions at 30 and platform cost 20: expected cost 180. At failure probability 0.30 it is 380. This does not estimate the intervention's causal effect; it only shows sensitivity to assumed residual risk.

The technical policy originally assumed perfect prevention at every triggered action. The extended test includes efficacy 0, 0.5 and 1 and capacity limits. When efficacy is zero, intervention expense can worsen total cost. Capacity zero recovers the no-intervention baseline. No future outcome is used to prioritize actions in this reference; capacity is consumed chronologically.

The legacy function named real_option is now explicitly a one-step assumed act/defer cost comparison. Its cost gap is not an option price. A proper staged-investment valuation would require states, information arrival, transition assumptions, admissible actions, discounting and financing constraints. These remain design options linked to Arctervix, not implemented findings.

Architecture, delivery and risk

Layers: approved observations -> quality checks -> time-aware forecast -> action feasibility -> economic scenario -> human decision -> outcome log. Keep a separate source/assumption ledger and controlled dashboard. Reuse maintenance systems and data tooling before building a platform; adapters and hosted services require separate access and confidentiality review.

Dashboard: data completeness, forecast error with sample size, candidate versus feasible actions, capacity used, action cost, residual failure exposure and assumed total cost. Distinguish predicted index, calibrated probability and actual event. No single aggregate score should conceal a safety constraint or unavailable spare part.

Material risks: leakage, weak labels, distribution shift, false confidence in efficacy, unavailable intervention resources and overlapping economic benefits. Owners are respectively model/data reviewer, operations specialist, model owner, maintenance owner, planner and finance reviewer. Stop a policy if it fails the same-case cost comparison or violates safety; retain the underlying technical knowledge for a better-scoped option.

Handoff and narrative

Deliver the data card, temporal evaluation plan, policy assumptions, scenarios, capacity/efficacy results, limitations and decision memo. The next real pilot requires an approved dataset and a defined operational decision, not mandatory user coursework.

Interview narrative: translate predictive analytics into an executable operating choice. Review exposed perfect-prevention assumptions and an overstated real-options label. The corrected package demonstrates how effectiveness and capacity can reverse the economic conclusion. It is a documented model review, not evidence of industrial savings or production deployment.

Validation evidence and reflection

Private synthetic study. Eight repository tests; twelve efficacy/capacity scenarios.

Efficacy	Capacity	Cost	Actions	Residual failures
0	0	540000	0	6
0	2	584000	2	6
0	10	760000	10	6
0	unlimited	1024000	22	6
0.5	0	540000	0	6
0.5	2	584000	2	6.0
0.5	10	670000	10	5.0
0.5	unlimited	754000	22	3.0
1	0	540000	0	6
1	2	584000	2	6
1	10	580000	10	4
1	unlimited	484000	22	0

No-intervention baseline cost: 540000. Synthetic technical cost units; not the EUR/period management case. Rolling index MAE: 0.039256; degraded-input MAE: 0.039728. This predicts an engineered index, not calibrated failure probability.

Decision and forecast interpretation

The original favorable-policy forecast holds only under its selected perfect-efficacy assumptions. Zero efficacy reverses the advantage; zero capacity reproduces no intervention. Preserve the negative result instead of presenting only the favorable scenario. The separate pre-recorded management-cost example is 180 versus 380 EUR/period at residual failure probabilities 0.10 and 0.30. This arithmetic does not identify treatment benefit. Continue a narrowly scoped decision study. Pause an operating recommendation until data quality, failure labels, efficacy and available intervention resources are established. A staged capital commitment requires a separate cash-flow and financing model.

Post-mortem

Removed the misleading option_value field: an absolute cost difference is not a real-option price. Added efficacy/capacity scenarios and rejected prediction-length mismatch instead of silently truncating observations. Costs, scores and probabilities retain distinct meanings. Handoff: DELIVERY-SPECIFICATION.md, tests and artifacts/management-review.json. No real asset, customer, industrial saving or production integration is claimed.